Dietary patterns, food groups, and telomere length in the Multi-Ethnic Study of Atherosclerosis (MESA)

Background Telomere length reflects biological aging and may be influenced by environmental factors, including those that affect inflammatory processes. Objective With data from 840 white, black, and Hispanic adults from the Multi-Ethnic Study of Atherosclerosis, we studied cross-sectional associations between telomere length and dietary patterns and foods and beverages that were associated with markers of inflammation. Design Leukocyte telomere length was measured by quantitative polymerase chain reaction. Length was calculated as the amount of telomeric DNA (T) divided by the amount of a single-copy control DNA (S) (T/S ratio). Intake of whole grains, fruit and vegetables, low-fat dairy, nuts or seeds, nonfried fish, coffee, refined grains, fried foods, red meat, processed meat, and sugar-sweetened soda were computed with responses to a 120-item food-frequency questionnaire completed at baseline. Scores on 2 previously defined empirical dietary patterns were also computed for each participant. Results After adjustment for age, other demographics, lifestyle factors, and intakes of other foods or beverages, only processed meat intake was associated with telomere length. For every 1 serving/d greater intake of processed meat, the T/S ratio was 0.07 smaller ($\beta \pm$ SE: -0.07 ± 0.03, $P = 0.006$). Categorical analysis showed that participants consuming ≥1 serving of processed meat each week had 0.017 smaller T/S ratios than did nonconsumers. Other foods or beverages and the 2 dietary patterns were not associated with telomere length. Conclusions Processed meat intake showed an expected inverse association with telomere length, but other diet features did not show their expected associations.